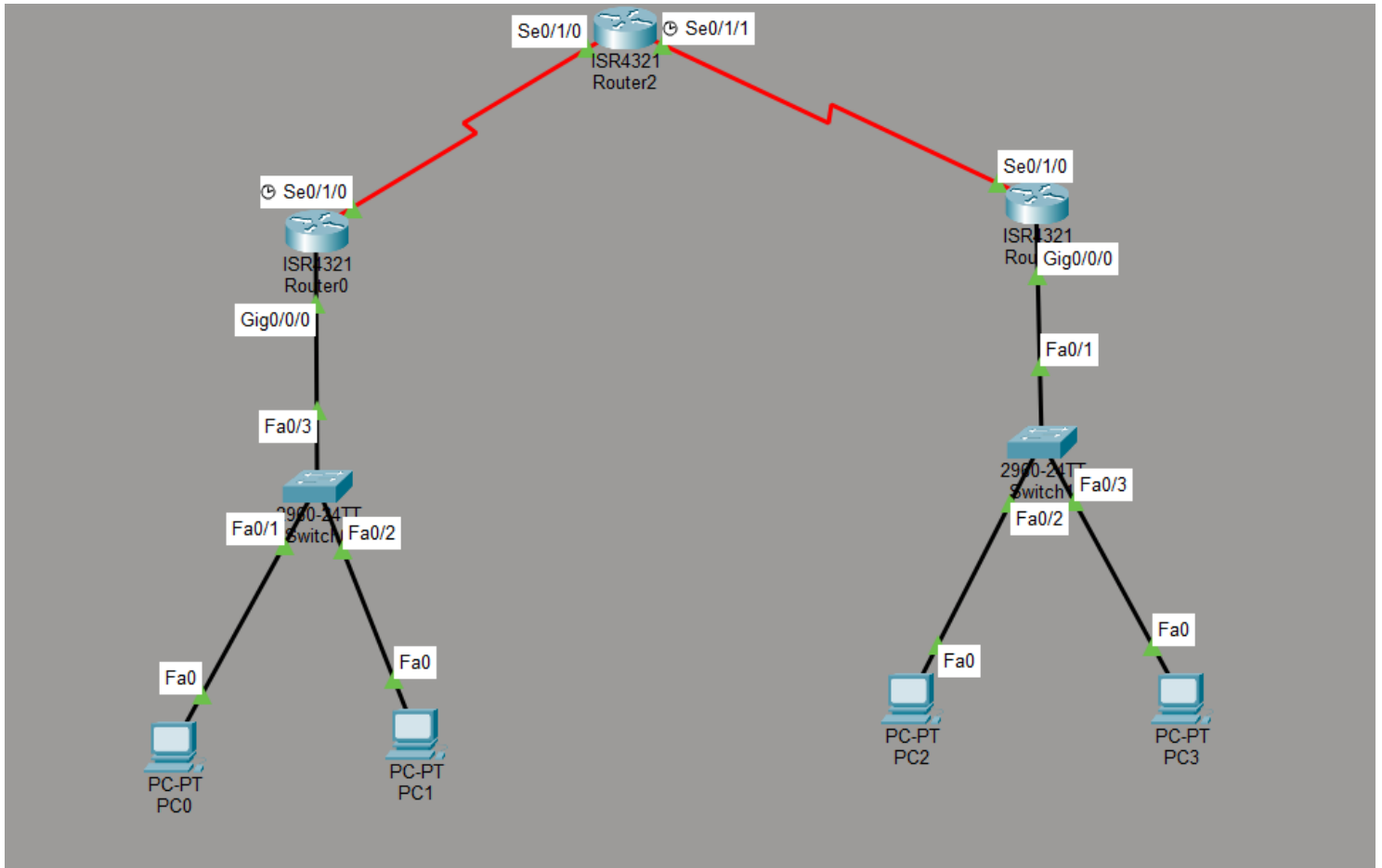


Q- Use Cisco Packet Tracer to simulate a WAN setup connecting two branch offices via a VPN tunnel and configure NAT on the edge routers so both branches can access the internet.



Router 0

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g 0/0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#int s 0/1/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#ip nat outside
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
Router(config-if)#exit
Router(config)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router(config)#ip address 10.0.0.4 255.255.255.252 10.0.0.2
^
% Invalid input detected at '^' marker.

Router(config)#ip route 10.0.0.4 255.255.255.252 10.0.0.2
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router(config)#access-list 100 permit ip 192.168.10.0 0.0.0.255 any
Router(config)#ip nat inside source list 100 interface serial 0/1/0 overload
Router(config)#do wr
Building configuration...
[OK]
Router(config)#access-list 110 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.6
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.6
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 110
Router(config-crypto-map)#exit
Router(config)#interface serial 0/1/0
Router(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.0.0.6      10.0.0.1      QM_IDLE        1058      0 ACTIVE

```

Router 1

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g 0/0/0/
      ^
% Invalid input detected at '^' marker.

Router(config)#int g 0/0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#int s 0/1/0
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#ip nat outside
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
Router(config-if)#
Router(config-if)#exit
Router(config)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router(config)#ip route 10.0.0.0 255.255.255.252 10.0.0.5
Router(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.5
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.5
Router(config)#do wr
Building configuration...
[OK]

```

```

Router(config)#access-list 100 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config)#access-list 100 permit ip 192.168.20.0 0.0.0.255 any
Router(config)#ip nat inside source list 100 interface serial 0/1/0 overload
Router(config)#do wr
Building configuration...
[OK]
Router(config)#access-list 110 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 110
Router(config-crypto-map)#exit
Router(config)#interface serial 0/1/0
Router(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#

```

Router 2

```

Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int s 0/1/1
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

Router(config)#int s 0/1/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router(config)#int loopback 0

Router(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

Router(config-if)#ip address 8.8.8.8 255.255.255.255
Router(config-if)#exit
Router(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.6
Router(config)#do wr
Building configuration...
[OK]
Router(config)#

```

PC 0 Ping

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=19ms TTL=126
Reply from 192.168.20.10: bytes=32 time=12ms TTL=126
Reply from 192.168.20.10: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 19ms, Average = 11ms

C:\>
```